

Solutions to Problems

30. Order of **decreasing** reactivity; for brevity, only the substituents are listed.

(a) $-\text{CH}_3 > -\text{CH}_2\text{Cl} > -\text{CH}_2\text{Cl} > -\text{CCl}_3$. $-\text{CH}_3 > -\text{CH}_2\text{Cl} > -\text{CH}_2\text{Cl} > -\text{CCl}_3$. carbon δ^+ , δ^+ , so the inductive effect becomes increasing deactivating.

(b) $-\text{O}^- \text{Na}^+ > -\text{OCH}_3 > -\text{OC}(=\text{O})\text{CH}_3$. Resonance activation

availability of oxygen's lone pair electrons, because they resonance donation to stabilize the cationic intermediate. oxygen atom is the best donor, the neutral ether oxygen in the ester worst because it is attached to a δ^+ carbon, electrons away from the ring.

(c) $-\text{CH}_2\text{CH}_3 > -\text{CH}_2\text{CCl}_3 > -\text{CH}_2\text{CF}_3 > -\text{CF}_2\text{CH}_3$. $-\text{CH}_2\text{CH}_3 > -\text{CH}_2\text{CCl}_3 > -\text{CH}_2\text{CF}_3 > -\text{CF}_2\text{CH}_3$. Inductive effect, distance to ring.

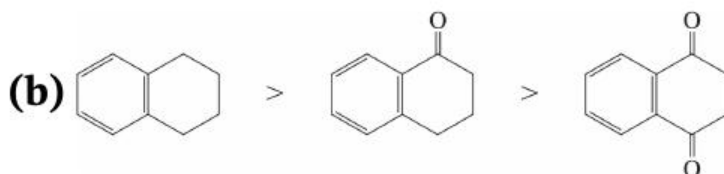
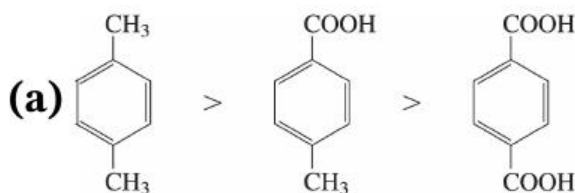
31. Compare the $-\text{CH}_2\text{Cl}-\text{CH}_2\text{Cl}$ substituent with the simplest similar one you know, the $-\text{CH}_3-\text{CH}_3$ group. We know from [Section 16-2](#) that the methyl group is activating because it is an electron-donating group by induction. The methyl group is also an *ortho,para*-directing

group, because its donating effect best stabilizes the carbocations (and therefore the transition states leading to the carbocations) resulting from substitution in the ortho and para positions.

Adding a chlorine atom to the methyl carbon reduces the inductive electron-donating capability of the group because chlorine is very electronegative and inductively electron-withdrawing. The effect is to make the $\text{—CH}_2\text{Cl—CH}_2\text{Cl}$ group slightly electron-withdrawing and, therefore, a weakly deactivating group. It also is less strongly *ortho,para*-directing, with now a significant amount of *meta*-substitution being observed, at least when nitration is the reaction being carried out. Table 16-2 provides a useful overview of nitrations of a variety of substituted benzenes. Notice that toluene (methylbenzene) nitrates mostly in the ortho position (58%). In (chloromethyl)benzene the amount of ortho nitration is cut almost in half (32%). It is likely that sterics play a role here: The $\text{—CH}_2\text{Cl—CH}_2\text{Cl}$ substituent is considerably more bulky than methyl, and therefore obstructs ortho substitution to a greater extent.

32. Activated: (c), (d), (e), (g)

33.

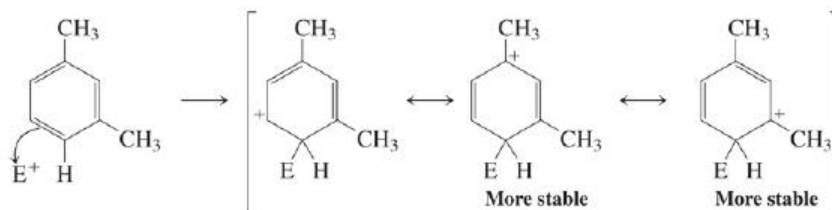


In each series of compounds, the benzene ring with two alkyl group substituents is most activated, the one with two carbonyl group substituents is most deactivated, and the structure with one of each is

intermediate.

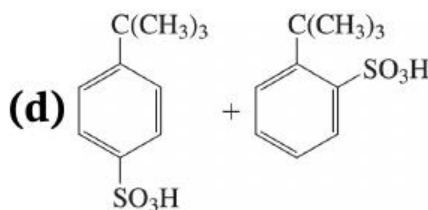
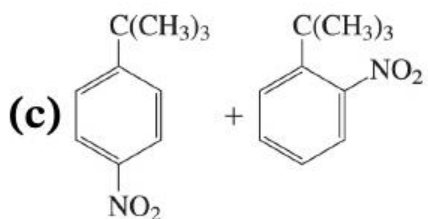
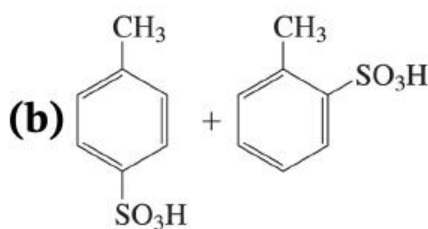
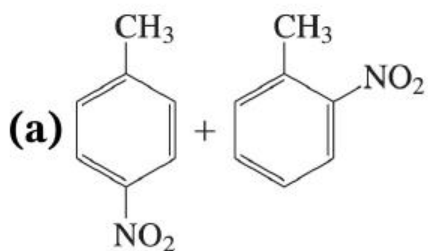
34. The two methyl substituents in 1,3-dimethylbenzene reinforce each other's activating and directing effects: They both direct subsequent electrophilic attack to positions 4 and 6 on the ring. For example, position 4 is ortho to the methyl group at C3 and para to the methyl group at C1, so it is doubly activated for subsequent substitution. In either 1,2- or 1,4-dimethylbenzene (*o*- or *p*-xylene) the two methyl groups direct to different carbons in the ring. For example, in 1,2-dimethylbenzene, the methyl group at C1 directs to positions 4 and 6, while the methyl group at C2 directs to positions 3 and 5.

Another way of looking at the situation is to notice that the intermediate cation resulting from attack at C4 on 1,3-dimethylbenzene has *two* resonance contributors in which the cationic carbon is methyl-substituted (hence, tertiary rather than secondary):



Therefore the energy of this cation is lower, and the activation barrier associated with its formation is reduced. Its formation is relatively fast. In contrast, none of the cations derived from electrophilic attack on either 1,2- or 1,4-dimethylbenzene are directly stabilized by more than one of the methyl groups. They are all higher in energy, and their formation requires surmounting a larger activation barrier.

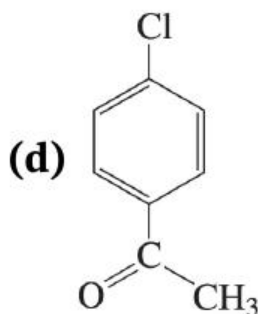
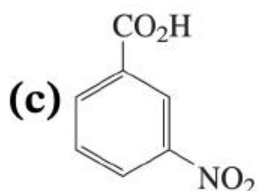
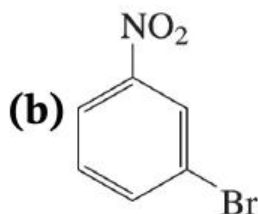
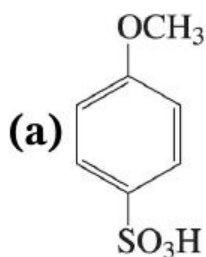
35.



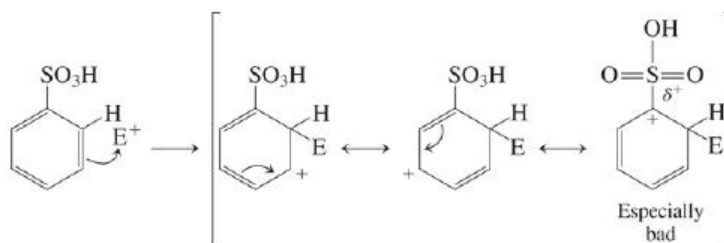
Changing from a small (methyl) group to a very bulky (1,1-dimethylethyl) substituent will strongly sterically inhibit ortho substitution. Reactions (c) and (d) give rise to almost entirely para-disubstituted products. Only reaction (a) gives ortho as the major product (text, Table 16.2). Both the methyl group and the nitronium cation are small, and there are two ortho positions competing against a single para site, so ortho wins. Reaction (b) gives mainly para: The electrophile in sulfonation is much bulkier, disfavoring ortho.

36. The position at which substitution occurs is determined by the

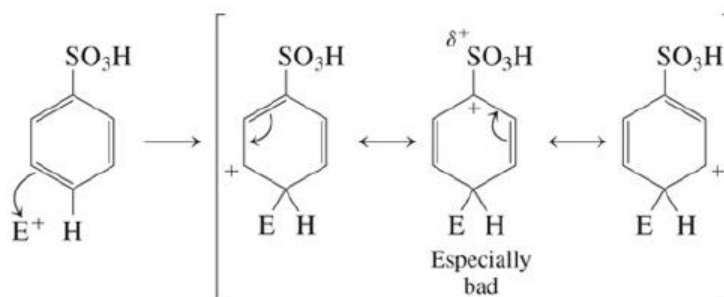
directing effect of the substituent that is already present in the starting compound. Methoxy and chloro are ortho, para-directing groups. Most of the time they direct incoming electrophiles to the para position, especially if the electrophile is somewhat bulky, as is the case in (a) and (d). Nitro and carboxylic acid are meta-directing.



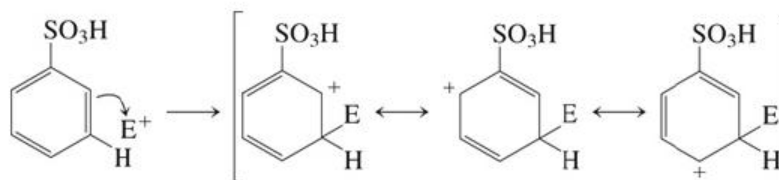
37. Ortho attack:



Para attack:



Meta attack:

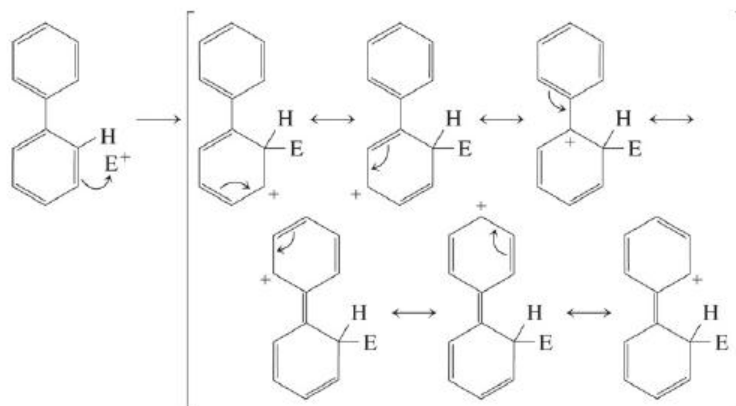


There are no resonance forms with $++$ adjacent to $\delta+\delta^+$ sulfur when attack occurs meta.

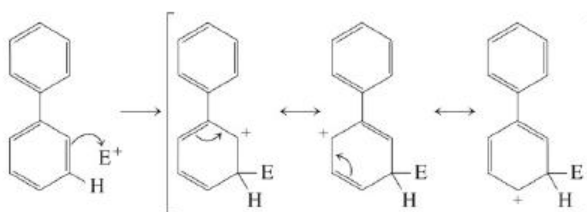
- 38.** Statement is correct. All meta-directors deactivate the entire ring by inductive electron withdrawal. Deactivation at the ortho and para positions is most intense as a result of resonance (see, for example, the answer to [Problem 37](#)). Meta substitution occurs only because the deactivation is felt least strongly at that position.
- 39.** Solve the problem by examining the structures, in particular the resonance forms, of the carbocation that is produced by attachment of a generic electrophile, E^+ , to each of the three possible positions on one of the benzene rings: ortho, meta, and para, with respect to the

other ring. Use principles developed in this chapter to evaluate the relative stabilities of these carbocations. The more stable carbocations should form faster and will determine the major product(s).

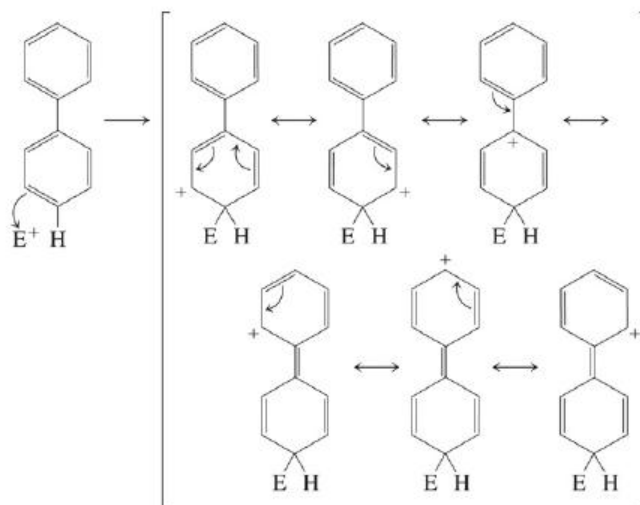
- (a)** Substitution ortho: attach the electrophile to either ring (they are equivalent), at the position adjacent to the bond connecting the two rings. Initially, we see the same three resonance forms that we can draw for substitution on benzene itself—the top three forms in the brackets below. However, having delocalized the positive charge to the position of attachment of the second ring (top right), we can continue to use the double bonds of this ring to delocalize further, giving us three more resonance forms (bottom three in brackets), for a total of six.



- (b)** Substitution meta: attach the electrophile one position further removed from the bond between the rings, the meta position. Draw the resonance forms for the carbocation. Notice now that the positive charge is no longer delocalized to the position of attachment to the second ring. This ring is incapable of providing additional resonance stabilization beyond that of the three original forms present in the ring on which the substitution took place.



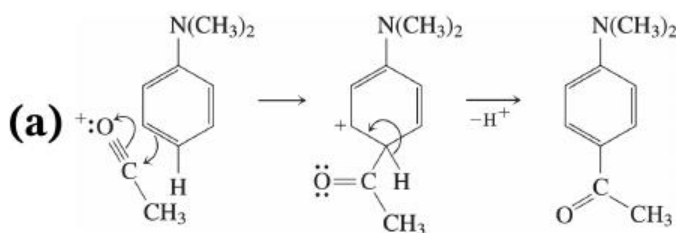
- (c) Substitution para: attach the electrophile and proceed as before. As was the case for substitution ortho, one of the three initial forms places the positive charge at the ring connection position, permitting three more resonance forms to participate in stabilizing the intermediate. (The second and third resonance forms in the top group of three have been written in reverse order—and the electron-pushing arrows modified accordingly—so that the form at the right leads naturally to the three forms in the bottom group of three.)



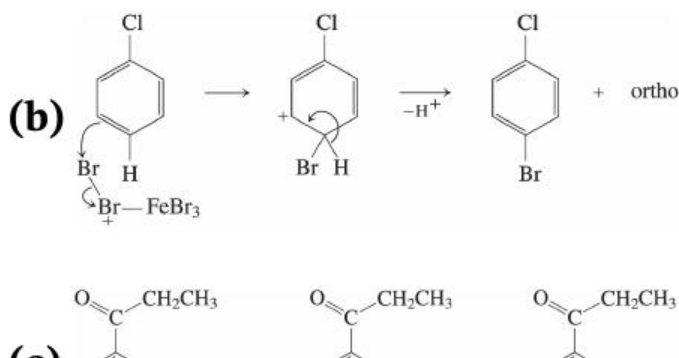
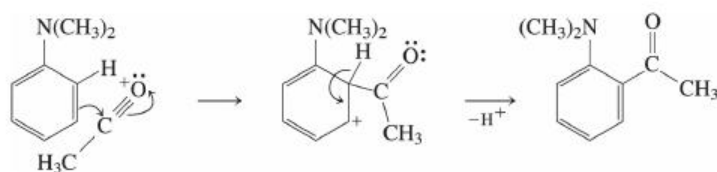
Additional resonance forms provide additional stabilization, and therein lies the answer. Substitution either ortho or para leads to a carbocation delocalized throughout both rings, with a total of six resonance forms, while meta substitution gives a cation limited to delocalization in only one of the rings, with only three contributing resonance forms. The meta cation is less stable, higher in energy, requires surmounting a higher

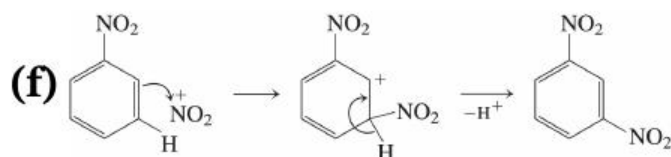
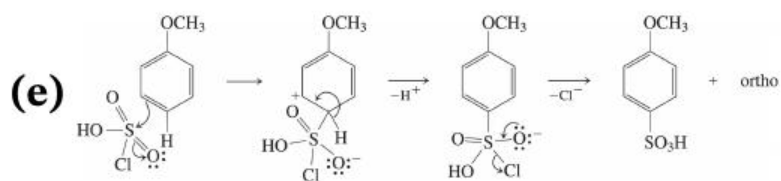
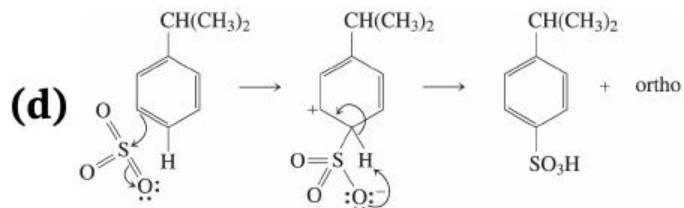
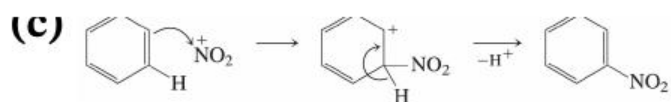
activation energy barrier to form, and therefore forms more slowly. The phenyl ring as a substituent is thus an ortho, para-directing group with respect to electrophilic aromatic substitution. It is also activating, again a consequence of the additional stabilization of the cation formed upon ortho or para substitution, relative to the cation formed from substitution on benzene itself.

10. Products and mechanisms [part (g)] are shown for each. Only the actual electrophile is shown; see the text for formation of each electrophile from the given precursors. Where both ortho and para products are formed, the para is generally the major one. The para mechanism is shown in all such cases; the ortho only for part (a).

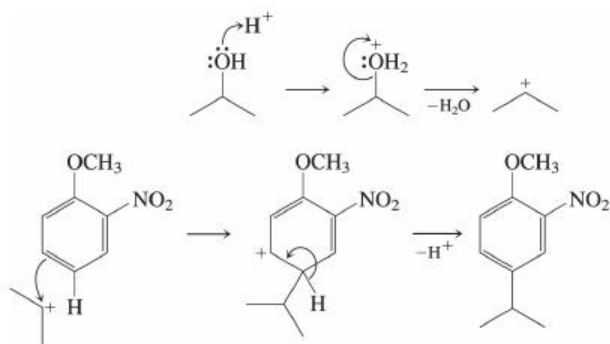


and





(g) Form the electrophile first:



11. **Note:** Write out the solutions for Problems 41 and 42 in full equation form for yourself.

- (a) Nitrobenzene + $\text{Br}_2, \text{FeBr}_3$ + $\text{Br}_2, \text{FeBr}_3$ (nitration of bromobenzene would give mostly ortho and para products);
- (b) iodobenzene + $\text{Br}_2, \text{FeBr}_3$; + $\text{Br}_2, \text{FeBr}_3$;
- (c) methoxybenzene + $\text{CH}_3\text{Cl}, \text{AlCl}_3$; + $\text{CH}_3\text{Cl}, \text{AlCl}_3$;
- (d) methoxybenzene + $\text{CH}_3\text{COCl}, \text{AlCl}_3$; + $\text{CH}_3\text{COCl}, \text{AlCl}_3$;
- (e) nitrobenzene + $\text{SO}_3, \text{H}_2\text{SO}_4$ + $\text{SO}_3, \text{H}_2\text{SO}_4$ or benzenesulfonic acid + $\text{HNO}_3, \text{H}_2\text{SO}_4$; + $\text{HNO}_3, \text{H}_2\text{SO}_4$;
- (f) toluene + $\text{HNO}_3, \text{H}_2\text{SO}_4$ + $\text{HNO}_3, \text{H}_2\text{SO}_4$ (Table 16-2: This reaction is somewhat unusual in giving mostly ortho rather than para product);
- (g) nitrobenzene + $\text{HNO}_3, \text{H}_2\text{SO}_4$; + $\text{HNO}_3, \text{H}_2\text{SO}_4$;
- (h) acetophenone + $\text{Cl}_2, \text{FeCl}_3$. + $\text{Cl}_2, \text{FeCl}_3$.

12. Note: These problems rely on the material in [Section 16-5](#). Review before you begin.

- (a) Nitrobenzene + 1. $\text{Br}_2, \text{FeBr}_3$ (brominates meta to nitro group), 2. Fe, HCl (reduces NO_2 to NH_2);
- (b) iodobenzene + $\text{SO}_3, \text{H}_2\text{SO}_4$ (sulfonates

para to iodine; blocks para position), 2. $\text{Br}_2, \text{FeBr}_3$ (brominates ortho to iodine), 3. $\text{H}^+, \text{H}_2\text{O}, \Delta\Delta$ (removes SO_3H group);

(c) methoxybenzene +1+1. $\text{CH}_2\text{CH}_2\text{CH}_2\text{COCl}, \text{AlCl}_3$ (brominates ortho to iodine), 2. $\text{H}^+, \text{H}_2\text{O}, \Delta\Delta$ (removes SO_3H group);
 $\text{CH}_2\text{CH}_2\text{CH}_2\text{COCl}, \text{AlCl}_3$ (**Caution:** Friedel-Crafts alkylation with 1-chlorobutane would give a rearranged product; use instead Friedel-Crafts acylation para to methoxy group), 2. H_2, Pd or Zn(Hg), HCl (reduce $\text{C}=\text{O}$ to CH_2);

(d) ethylbenzene +1+1. $\text{CH}_3\text{COCl}, \text{AlCl}_3$ (acetylates para to ethyl group), 2. $\text{CrO}_3, \text{H}_2\text{SO}_4, \text{H}_2\text{O}$ (oxidizes original ethyl group to acetyl);

(e) aniline +1+1. $\text{CH}_3\text{COCl}, \text{pyridine}$ (protects amino group as amide), 2. HNO_3 (nitrates para), 3. $\text{H}^+, \text{H}_2\text{O}, \Delta\Delta$ and 4. $-\text{OH}, \text{H}_2\text{O}$ (frees NH_2 group);

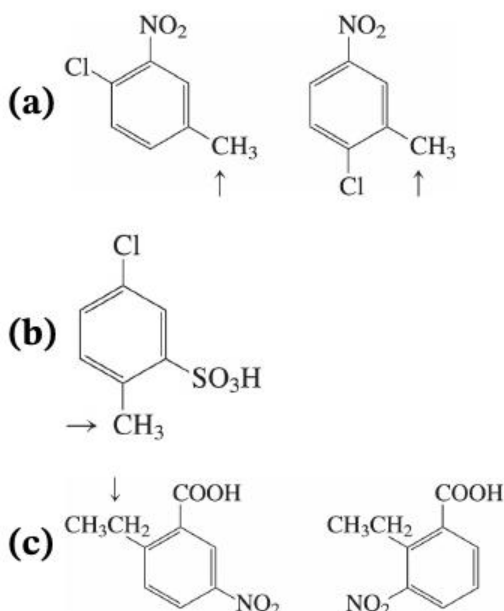
(f) ethylbenzene + $\text{SO}_3, \text{H}_2\text{SO}_4$ (sulfonates para to iodine; blocks para position), 2. $\text{Br}_2, \text{FeBr}_3$ (brominates ortho to ethyl group), 3. $\text{H}^+, \text{H}_2\text{O}, \Delta\Delta$ (removes SO_3H group) (**Caution:** You cannot begin with bromobenzene, because after sulfonation para you cannot do the necessary subsequent Friedel-Crafts reaction. Friedel-Crafts reactions do not normally proceed when a *meta*-

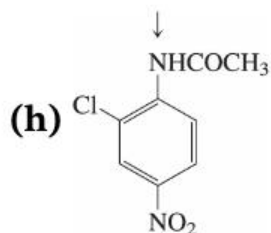
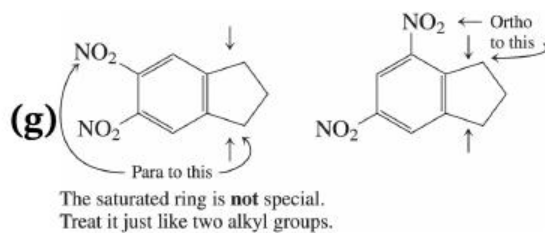
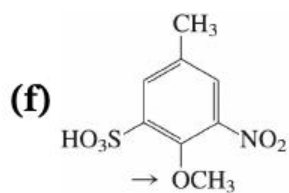
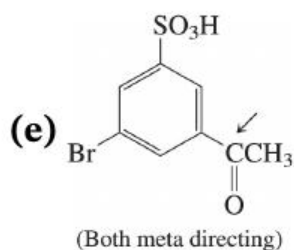
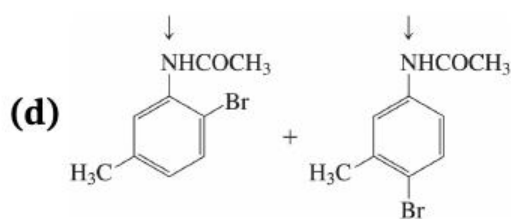
directing group, such as SO_3H , SO_3Na , is present);

(g) nitrobenzene +1+1. Fe, HCl (Fe, HCl (reduces NO_2 to NH_2), 2. CH_3COCl , CH_3COCl , pyridine (protects amino group as amide), 3. HNO_3 (HNO_3 (nitrates para), 4. H^+ , H_2O , H^+ , H_2O , Δ and 5. $-\text{OH}$, H_2O (H_2O (frees NH_2 group), 6. $\text{CF}_3\text{CO}_3\text{H}$ ($\text{CF}_3\text{CO}_3\text{H}$ (oxidizes NH_2 to NO_2);

(h) acetophenone +1+1. $\text{SO}_3, \text{H}_2\text{SO}_4$, $\text{SO}_3, \text{H}_2\text{SO}_4$, 2. H_2, Pd . H_2, Pd .

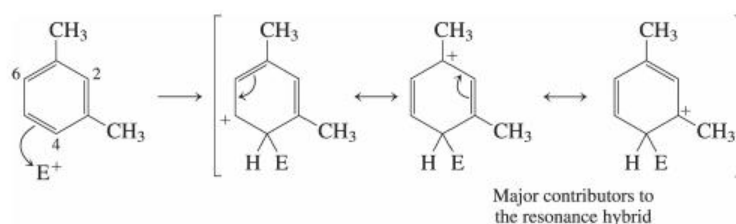
13. Orientation of reaction is determined by more activating (or less deactivating) substituent (marked in each case below). Again, the para product can be expected to predominate where a choice of ortho or para exists.





(i) No reaction. Friedel-Crafts reaction does not occur on rings containing meta-directing groups: If there are no strongly activating groups present to counter them, the ring is too deactivated.

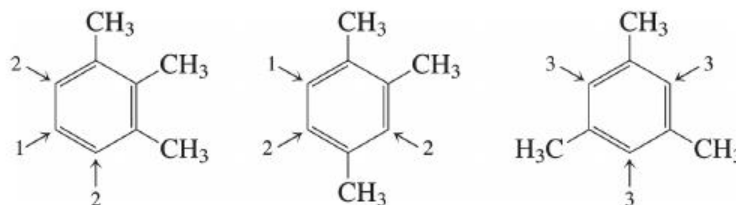
- (a) Activation effects are additive. 1,3-Dimethylbenzene (*m*-xylene) differs from its 1,2 and 1,4 isomers because it is the **only** one that contains ring positions activated by **both** methyl substituents. Electrophilic substitution at C2, C4, and C6 (which is equivalent to C4) gives intermediate cations whose positive charges are delocalized to two methyl-substituted positions around the ring. For attack at C4, you have



The cations are more stable, the transition states leading to them are lower in energy, and they form faster. Substitution mainly occurs at C4/C6 (C2 is a sterically hindered position because of the methyls on either side).

You should check for yourself that electrophilic substitution on the ortho and para isomers cannot lead to such a doubly stabilized intermediate cation.

- (b) Three methyl groups, but the principle to be applied is the same: Look for positions that are doubly or triply activated. In each structure count up the number of methyls ortho or para to each open position.



The last should be the most reactive toward electrophiles: Each vacant position is activated by being ortho or para to **all**

three methyl groups. By the way, the reactivity difference between the 1,3,5 isomer and the other two is quite large, about a factor of 200 in rate.

- 15.** Friedel-Crafts reactions are understood to be followed by H^+ , H_2O work-up.

(a) 1. $\text{CH}_3\text{CH}_2\text{Cl}$, AlCl_3 ; $\text{CH}_3\text{CH}_2\text{Cl}$, AlCl_3 ; 2. CH_3COCl , AlCl_3 ; CH_3COCl , AlCl_3

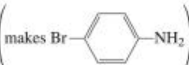
(b) 1. HNO_3 , H_2SO_4 ; HNO_3 , H_2SO_4 ; 2. Cl_2 , FeCl_3 ; Cl_2 , FeCl_3

(c) 1. CH_3COCl , AlCl_3 ; CH_3COCl , AlCl_3 ; 2. SO_3 , H_2SO_4
 SO_3 , H_2SO_4
 Can't do it in the reverse order. The Friedel-Crafts reaction fails with the SO_3H group there.

(d) 1. HNO_3 , H_2SO_4 ; HNO_3 , H_2SO_4 ; 2. HCl , Zn(Hg) ; HCl , Zn(Hg) ; 3. SO_3 , H_2SO_4 ; SO_3 , H_2SO_4 , heat; 4. $\text{CF}_3\text{CO}_2\text{H}$, $\text{CF}_3\text{CO}_2\text{H}$

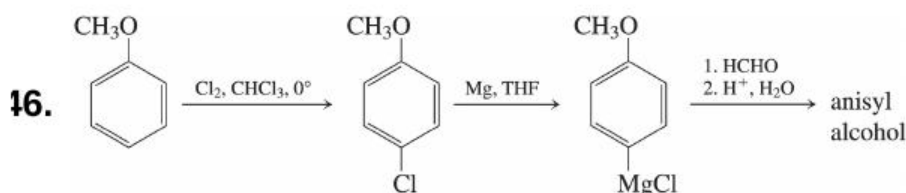
(e) 1. Cl_2 , FeCl_3 ; Cl_2 , FeCl_3 ; 2. excess conc. HNO_3 , H_2SO_4 , HNO_3 , H_2SO_4 , $\Delta\Delta$

(f) 1. Br_2 , FeBr_3 ; Br_2 , FeBr_3 ; 2. HNO_3 , H_2SO_4 , HNO_3 , H_2SO_4 , separate para from ortho;

(g) 3. HCl , Zn(Hg) ; HCl , Zn(Hg) (makes ); 4. Cl_2 , CHCl_3 , 0°C to 0°C (chlorinates once, ortho to NH_2); see [Section 16-3](#); 5. $\text{CF}_3\text{CO}_2\text{H}$, $\text{CF}_3\text{CO}_2\text{H}$

(h) 1. $\text{Br}_2, \text{FeBr}_3; \text{Br}_2, \text{FeBr}_3$; 2. $\text{SO}_3, \text{H}_2\text{SO}_4; \text{SO}_3, \text{H}_2\text{SO}_4$
(blocks para); 3. $\text{Cl}_2, \text{FeCl}_3; \text{Cl}_2, \text{FeCl}_3$; 4. $\text{H}_2\text{O}, \text{H}_2\text{O}, \Delta\Delta$

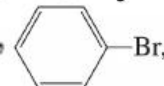
(i) 1. $\text{CH}_3\text{Cl}, \text{AlCl}_3; \text{CH}_3\text{Cl}, \text{AlCl}_3$; 2. $\text{SO}_3, \text{H}_2\text{SO}_4; \text{SO}_3, \text{H}_2\text{SO}_4$;
3. excess $\text{Br}_2, \text{FeBr}_3; \text{Br}_2, \text{FeBr}_3$; 4. $\text{H}_2\text{O}, \text{H}_2\text{O}, \Delta\Delta$



17. Sulfonation of salicylic acid should do nicely.

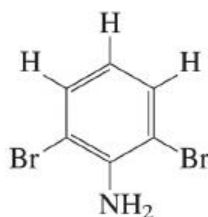
18. The synthesis of URB597 was extremely challenging, more so for the positions of its substituents than for their structure. Both groups are located in meta positions relative to the bond between the rings.
[Problem 39](#) revealed that phenyl rings as substituents are ortho, para directing. Therefore electrophilic aromatic substitution reactions performed on biphenyl cannot serve as an entry to the preparation of this important compound.

19.

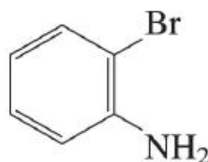
(a) NMR shows two sets of benzene hydrogens in a 2 : 32 : 3 integration ratio, and the IR bands at 685 and 735 cm^{-1} indicate a monosubstituted benzene. This analysis leads to the only logical answer, bromobenzene, $\text{C}_6\text{H}_5\text{Br}$, , which is made from benzene + Br_2 and FeBr_3 .

(b) NMR shows three benzene hydrogens, a 1 H triplet and a 2 H doublet; what you would expect for

three hydrogens in adjacent positions around a ring, with the end ones equivalent: HC—CH—CH. HC—CH—CH. What else is there? A 2 H₂ H lump at $\delta=4.5$ ppm together with two bands in the IR at 3382 and 3478 cm⁻¹ strongly suggest an NH₂ group. How about an atom count? You're up to six carbons for the benzene ring, five H and an N. The only formula that will fit is C₆H₅Br₂N. Putting the NH₂ on the benzene ring and then attaching the two Br's one each side of it fits the data: 2,6-dibromoaniline is the answer.



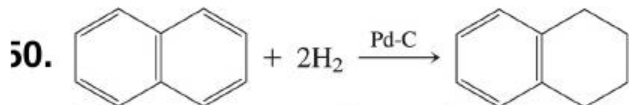
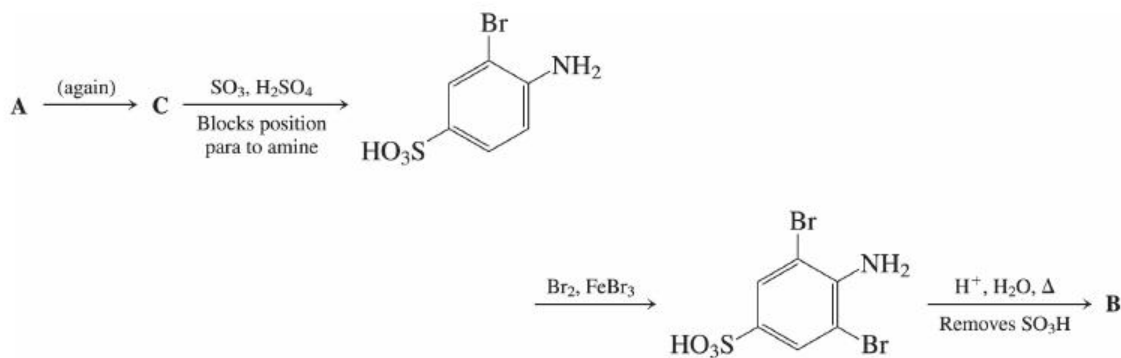
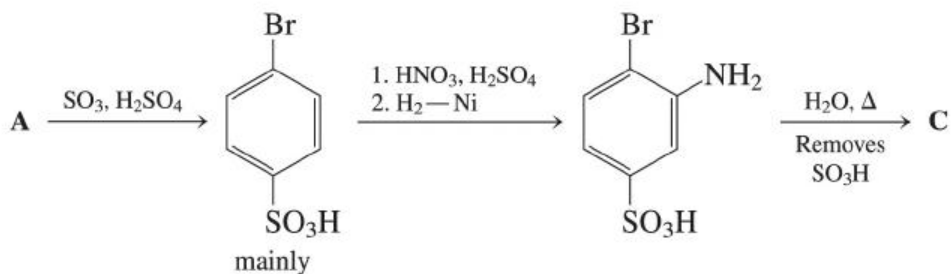
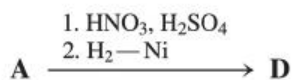
- (c) The NH₂ is again present, but the NMR now shows four benzene hydrogens in a complicated pattern. The IR helps: The band at 745 cm⁻¹ is in the range for ortho disubstituted rings. 2-Bromoaniline is the answer: C₆H₆BrN.



- (d) Similar, but the NMR (two doublets, each 2 H) and the IR (820 cm⁻¹) indicate that this is the para isomer of C₆H₅BrN: 4-bromoaniline:

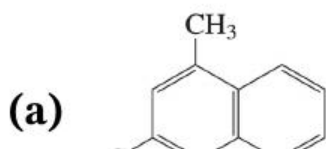


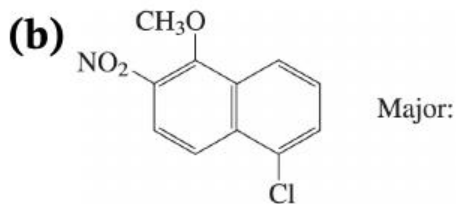
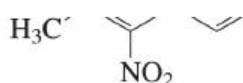
Synthesis. (Assume any ortho + para mixtures can be readily separated to give the para product in good yield.)



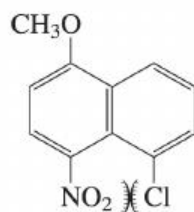
Although aromatic, naphthalene does undergo some addition reactions like this one. Addition occurs in such a way that one aromatic benzene ring is left fully intact.

51. Reaction should occur on the most activated (or least deactivated) ring, directed by the groups present.

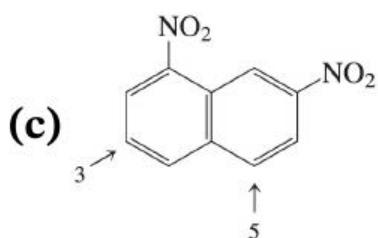




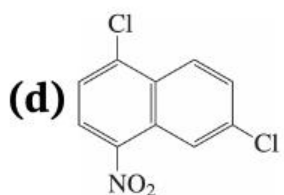
Major:



is sterically hindered



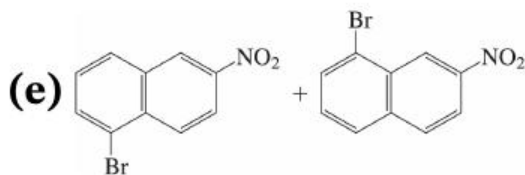
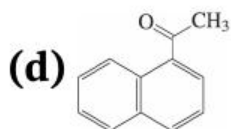
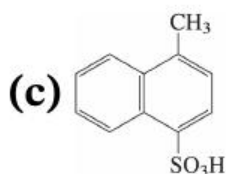
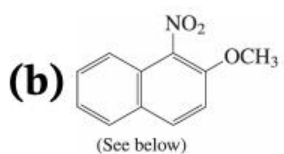
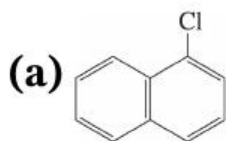
Two choices: C3C3 (meta to C1 NO₂)^{C1 NO₂} and C5C5 (meta to C7 NO₂)^{C7 NO₂}. If all other factors are equal, substitution next to a ring fusion is preferred because one benzene ring is left intact in the intermediate. (See structures for (b) of the solution to [Problem 46](#).) So, nitration occurs at C5.C5.



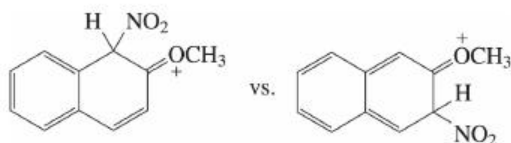
Reasons: para to a C1,C1, next to a ring

fusion, and relatively unhindered

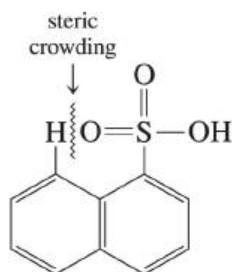
52.



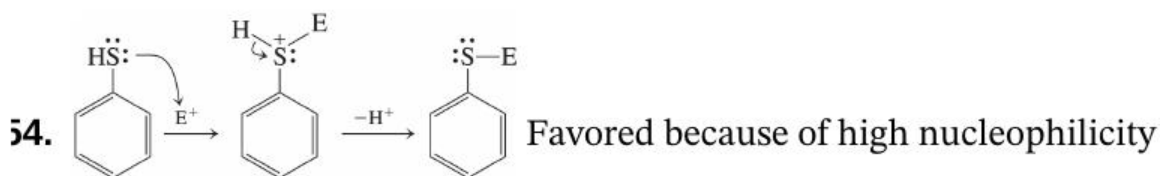
In (b), substitution at C1 is favored because the most important resonance form for the intermediate carbocation has an intact 6- π 6- π electron benzene ring (see structure, below left). The cation from substitution at C3 does not have this (below, right).



53. 1-Naphthalenesulfonic acid is the *kinetic* product: It forms faster because the cationic intermediate leading to its formation is more stable, and therefore is produced via a pathway with a lower activation barrier. However, 1-naphthalenesulfonic acid itself is less stable than 2-naphthalenesulfonic acid because in 1-naphthalenesulfonic acid the sulfonic acid group is sterically crowded by the hydrogen atom at C8:C8 :



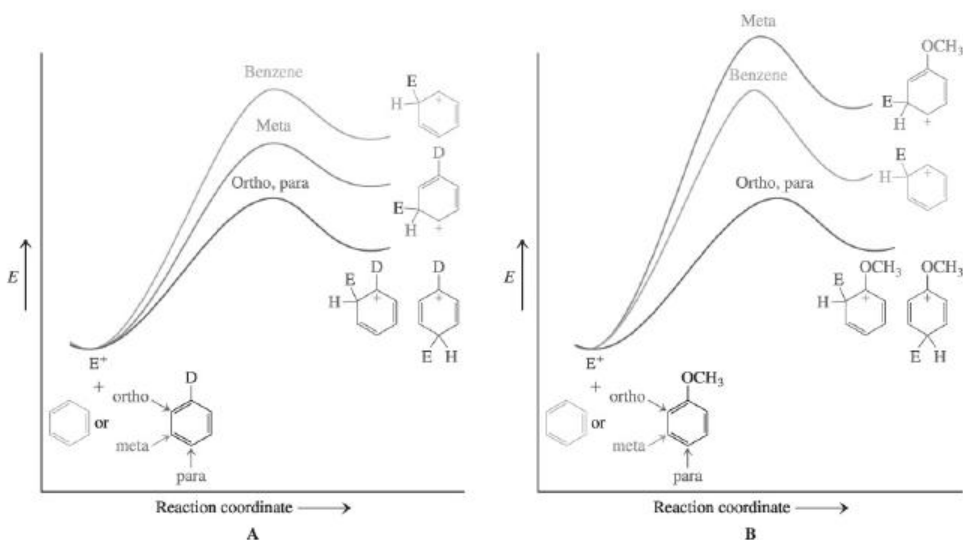
This fact would be completely irrelevant if not for the fact that *sulfonation is reversible*. So, at higher temperature desulfonation from C1C1 of the kinetic product can take place and (slower) sulfonation at C2C2 can occur, ultimately leading to build-up of the more stable product, 2-naphthalenesulfonic acid.



of sulfur atom

55.

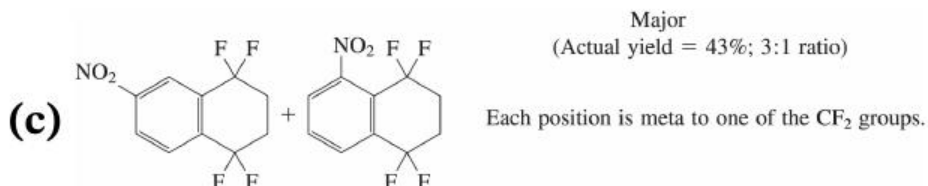
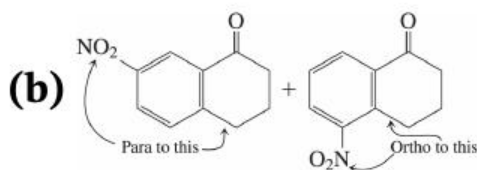
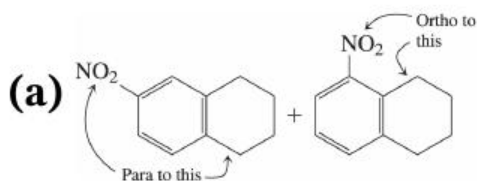
- (a) The methoxy group is *inductively electron withdrawing* because of the electronegative oxygen atom. The overwhelming resonance effect strongly activates the ortho and para positions, but it is **not directly felt** at the meta positions (see related resonance forms for aniline, [Section 16-3](#)). The deactivating inductive effect wins at the meta positions.
- (b) Based on the observed reactivity of anisole, it must be the case that the activation barrier for meta attack is higher than that for benzene itself, while the barrier for attack ortho or para is lower. So the graph should look like this:



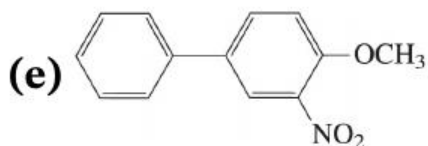
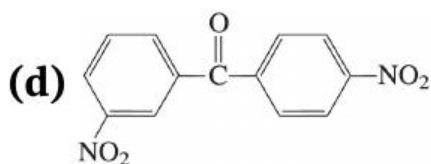
Schore, *Study Guide and Solutions Manual for Organic Chemistry*, 8e, © 2018 by W. H. Freeman and Company

56. See the answer to [Problem 43\(g\)](#). The saturated six-membered ring is nothing special. You can view it just like two alkyl substituents in adjacent positions on the benzene ring. For example, the starting

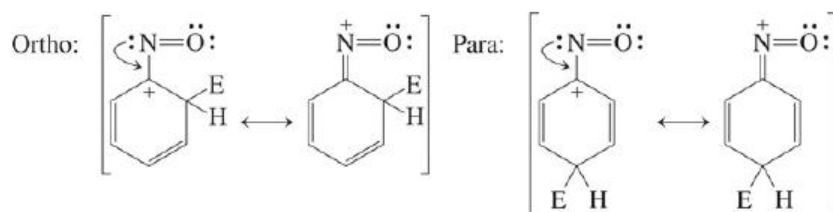
hydrocarbon in (a) can be treated just as if it were 1,2-dimethylbenzene (*o*-xylene).



In (d) and (e), monosubstitution occurs in the more activated or less deactivated ring.

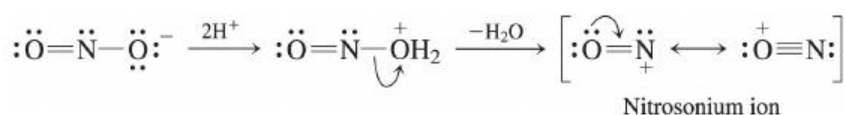


57. Lewis structure: $\text{--}\ddot{\text{N}}=\ddot{\text{O}}\text{:}$. The lone pair on NN will favor ortho and para substitution through the following resonance forms:

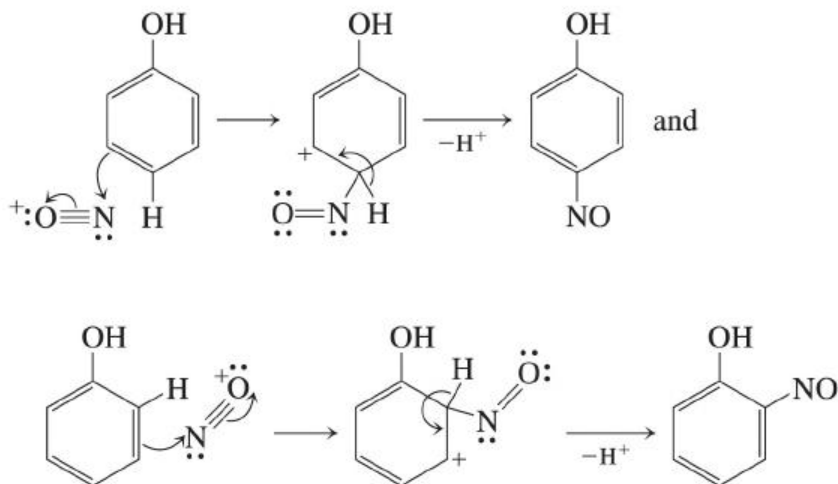


However, the —NO—NO group is inductively electron withdrawing. As is the case with halogen substituents, this deactivating inductive effect is on the average stronger than the resonance effect, so nitrosobenzene is deactivated overall, even though substitution is preferred (by the resonance effect) at the ortho and para positions.

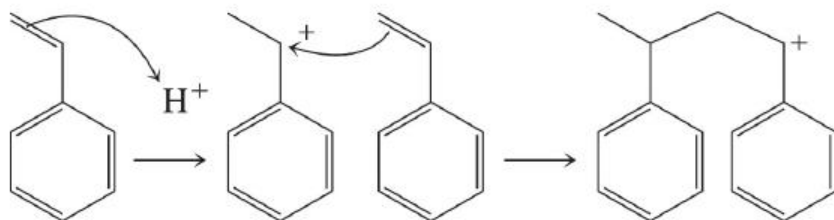
58. The electrophile:



Then

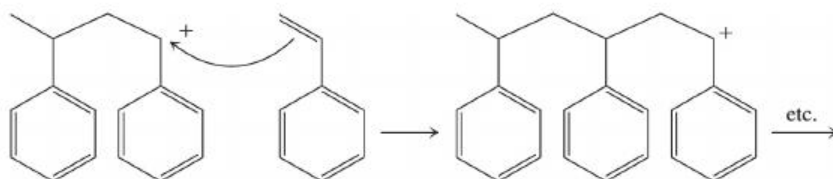


59.

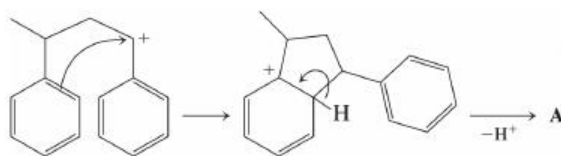


At this point the system has two options: add another styrene and proceed with polymerization, or close a ring by intramolecular Friedel-Crafts reaction.

Polymerization:



Friedel-Crafts:



30. (c)

31. (b)

32. (c)

33. (c)

34. (d)

17

Aldehydes and Ketones: The Carbonyl Group

Congratulations! You have finally gotten to the first of several chapters that examine the chemistry of carbonyl compounds: the most important ones in organic chemistry. Why are they so important? They are extremely versatile in carbon–carbon bond formation, and, therefore, synthesis. Carbonyl compounds contain an electrophilic carbonyl carbon (which you already know about from [Chapter 8](#)) as well as a potentially nucleophilic carbon next to it (which you will learn about shortly). This “double-barreled” functional capability is unique among the simple compound classes.

The importance of carbonyl compounds extends to biological chemistry as well, where the carbonyl group plays a central role in biochemical synthesis of naturally occurring molecules.

Outline of the Chapter

[17-1 Naming the Aldehydes and Ketones](#)

[17-2 Structural and Physical Properties](#)

[17-3 Spectroscopy](#)

[17-4 Preparation of Aldehydes and Ketones: A Review](#)

17-5 Reactivity of the Carbonyl Group: Mechanisms of Addition

One of two major reaction patterns for carbonyl compounds.

17-6, 17-7, 17-8 Addition of Water and Alcohols to Aldehydes and Ketones

Hydrates, hemiacetals, acetals, and protecting groups.

17-9 Addition of Amines to Aldehydes and Ketones

17-10 Deoxygenation of the Carbonyl Function

Reduction of $\text{C}=\text{O}$ to CH_2 . $\text{C}=\text{O}$ to CH_2 .

17-11, 17-12 Addition of Carbon Nucleophiles to Aldehydes and Ketones

With mechanistic details and synthetic applications.

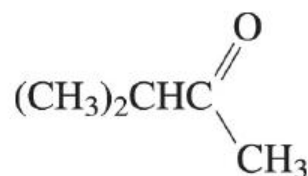
17-13, 17-14 Oxidations of Aldehydes and Ketones

Keys to the Chapter

17-1. Naming the Aldehydes and Ketones

Most of the material in these first sections is of a relatively routine nature, so only a few points of special interest, or with special implications will be mentioned.

Nomenclature of carbonyl compounds presents a bit of a problem in that several alternative names may be possible for almost any compound.



For example, one can give any of several names to the structure above: 3-Methyl-2-butanone (IUPAC) and isopropyl methyl ketone (common) are just two that are still in current use. Be prepared for some variety, especially in common names! The old (and I mean **old**) naming system for phenyl ketones is especially entertaining. That for $\text{C}_6\text{H}_5\text{COR}$ $\text{C}_6\text{H}_5\text{COR}$ was derived by combining the (common) name of the carboxylic acid RCO_2H RCO_2H (dropping the final -ic acid-*ic acid* and adding an *o* if one wasn't already there) with the suffix -phenone. So, $\text{C}_6\text{H}_5\text{COCH}_3$ $\text{C}_6\text{H}_5\text{COCH}_3$ is acet(ic acid)+o+phenone=acetophenone; acet(ic acid)+o + phenone = acetophenone; $\text{C}_6\text{H}_5\text{CO}(\text{CH}_2)_4\text{CH}_3$ $\text{C}_6\text{H}_5\text{CO}(\text{CH}_2)_4\text{CH}_3$ is valer(ic acid)+o+phenone=valerophenone; valer(ic acid)+o + phenone = valerophenone; $\text{C}_6\text{H}_5\text{COC}_6\text{H}_5$ $\text{C}_6\text{H}_5\text{COC}_6\text{H}_5$ is benzo(ic acid)+phenone=benzophenone; benzo(ic acid)+phenone = benzophenone; and so on. No, it doesn't make much sense to me, either, but that's what people called them then, and still often do now.

17-2. Structural and Physical Properties

The polarized carbonyl group is the key to the physical and chemical properties of carbonyl compounds. Although they have polarizations comparable to those found in haloalkanes, carbonyl compounds have a negatively polarized oxygen with lone pairs capable of hydrogen bonding to protic solvents. They are therefore much more water soluble than haloalkanes, for instance. The carbon–oxygen double bond is also stronger than the carbon–carbon double bond in alkenes. One consequence of the relatively strong $\text{C}=\text{O}$ bond is its tendency to form whenever possible. Whereas additions to $\text{C}=\text{C}$ bonds are usually quite exothermic, many additions to $\text{C}=\text{O}$ bonds are not and, indeed, are often reversible, with equilibrium constants near 1. Later you will see a number of reactions that generate carbonyl groups in ways that, at first glance, seem rather surprising.

17-3. Spectroscopy

A few noteworthy points are brought up here. Infrared spectroscopy is very useful for characterization of carbonyl compounds: The band for the $\text{C}=\text{O}$ stretch is very intense and located in a region of the spectrum (usually $1690\text{--}1750\text{ cm}^{-1}$) that is free from strong absorptions of other functional groups. The precise location of this peak is also useful in establishing the nature of the groups attached to the carbonyl carbon. ^{13}C NMR spectroscopy exhibits signals near $\delta = 200\text{ ppm}$ for the carbonyl carbon, and the aldehydic (also called ‘formyl’) hydrogen ($-\text{CHO}$) resonates in the range $\delta = 9.5\text{--}10.0\text{ ppm}$ in the ^1H NMR spectrum.

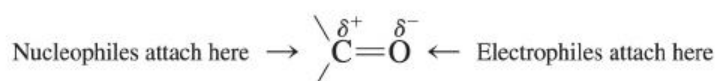
17-4. Preparation of Aldehydes and Ketones: A Review

The number of methods that exist to synthesize carbonyl compounds is impressive to the point of being intimidating. But don’t be put off by this

situation. Each reaction is presented in an appropriate, logical context. This section mainly reiterates carbonyl syntheses that you've seen before, with a few new examples to help reinforce those original presentations.

17-5. Reactivity of the Carbonyl Group: Mechanisms for Addition

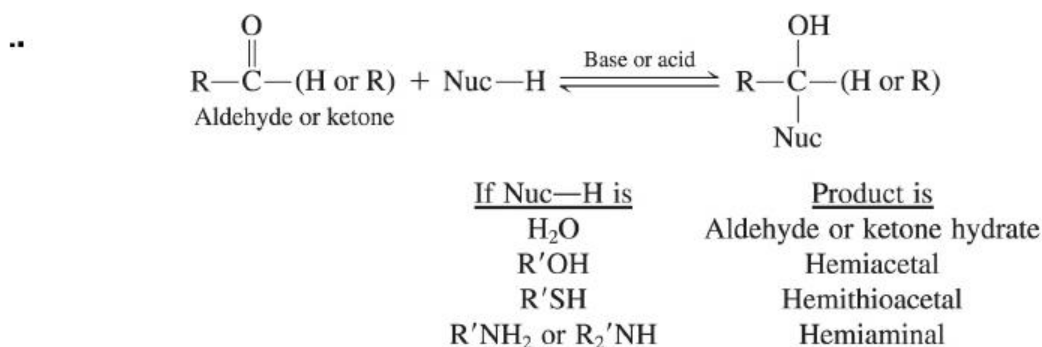
This chapter concentrates solely on one type of reaction: addition across the carbon – oxygen double bond of a carbonyl group. Carbonyl additions are polar reactions that exactly follow pathways that would be expected by electrostatics.



The two main mechanisms described in this section are distinguished by the order of addition. Strong nucleophiles, $\text{Nuc}:-\text{Nuc}^-$ (which may be added directly or formed by the reaction of a base with $\text{Nuc}-\text{HNuc}-\text{H}$), add to the carbonyl carbon first, followed (usually) by protonation of oxygen (which may be a separate step). Conversely, addition of weaker nucleophiles, especially neutral $\text{N}^{\text{uc}}\text{H}$, $\ddot{\text{N}}^{\text{uc}}\text{H}$, is helped by prior protonation of the carbonyl oxygen to give the highly electrophilic $\begin{array}{c} \diagup \quad \diagdown \\ \text{C}=\text{O}^+ \text{H} \\ \diagdown \quad \diagup \end{array}$ group. Whichever way the reaction proceeds in any given case, its scope is very broad, and many useful types of addition products are known. These are the subjects of the remaining sections of the chapter.

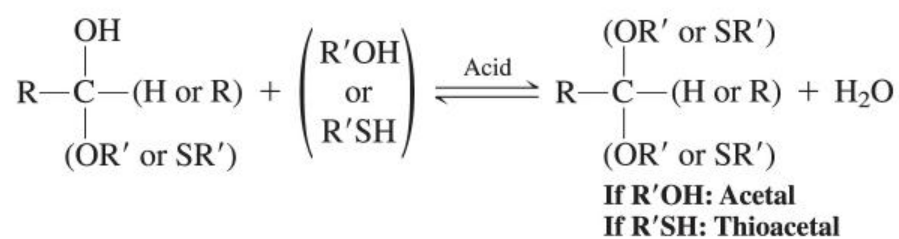
17-6 through 17-9. Addition of Water, Alcohols, and Amines to Aldehydes and Ketones

There are actually two fundamentally different types of reactions in these sections. They correspond to the two stages of the reactions under discussion. The first is the reversible addition of a nucleophile to a carbonyl group, which can generally be catalyzed by either base or acid.



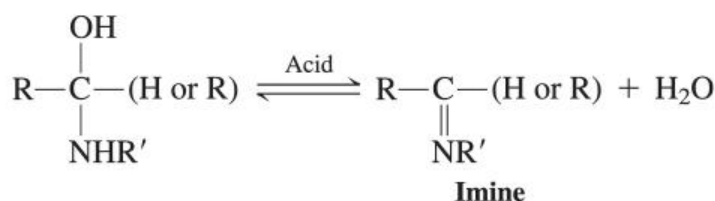
The second is really *carbocation chemistry*. When Nuc—HNuc—H is R'OH, R'OH, or R'SH, R'SH, the product in the reaction shown above may have its OH group replaced by a second Nuc. This reaction occurs by the S_N1 mechanism and leads to relatively stable acetal or thioacetal products. As the text section shows, these latter two kinds of compounds are resistant to attack by many kinds of basic or nucleophilic reagents, such as RLi, RLi, Grignards, and hydrides. Conversion of an aldehyde or ketone C=O to an acetal or thioacetal is a handy way of protecting it when you need to react some other functional group in a molecule with a strong base or nucleophile.

2a.

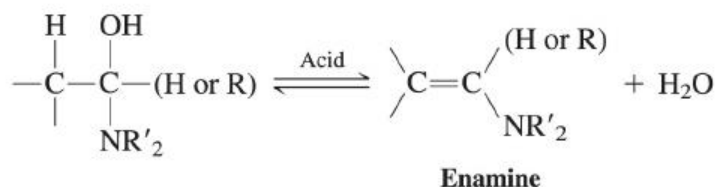


When Nuc—HNuc—H is ammonia or a primary amine, R'NH₂, R'NH₂, further reaction of the hemiaminal gives an **imine**, containing a carbon – nitrogen double bond. Alternatively, when the nucleophile is R'₂NH, R'₂NH, an **enamine** results.

2b.



2c.



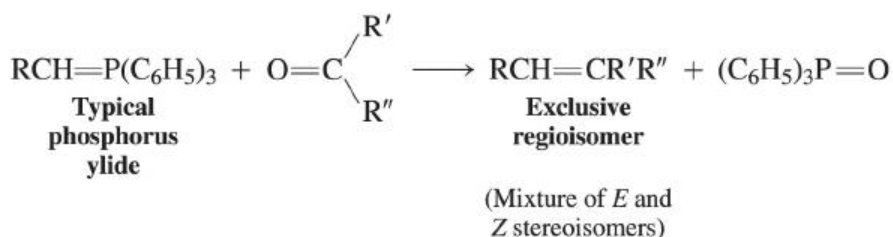
For all practical purposes, acid-catalyzed reaction of an aldehyde or a ketone with any of the nucleophiles listed in reaction 1 will continue straight through to the product of reactions 2a, 2b, and 2c, as the case may be. Conversely, acid-catalyzed hydrolysis of any of the latter products will proceed back all the way to the original aldehyde or ketone and release the free original nucleophile.

17-10. Deoxygenation of the Carbonyl Function

The reduction of $\text{C}=\text{O}$ to CH_2CH_2 may be achieved in three ways: Raney nickel desulfurization of thioacetals ([Section 17-8](#)), Clemmensen reduction ([Section 16-5](#)), and, as described here, Wolff-Kishner reduction. Because there are many ways to make carbonyl compounds, and carbonyl compounds are great places to start for making new carbon – carbon bonds, you will find that carbonyl groups are often present when complex molecules are made from simpler ones. This will be especially evident in [Chapters 18 – 20](#) and [23](#). Deoxygenation will be useful if you need to get rid of a carbonyl group that has been used to construct bonds in a molecule but is not wanted in the final product. The example of Friedel-Crafts acylation followed by deoxygenation to give an alkylbenzene is a case in point.

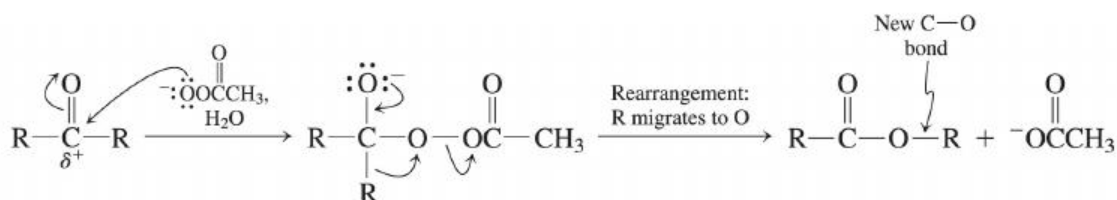
17-11 and 17-12. Addition of Carbon Nucleophiles to Aldehydes and Ketones

Grignard reagents and alkyllithiums are examples of highly reactive “carbanionic” reagents (i.e., they behave like carbanions). In this section less highly energetic carbanionic reagents are introduced. Cyanide ion is the simplest, and the products of its addition to aldehydes and ketones, cyanohydrins, have some specialized synthetic utility. Ylide reagents containing phosphorus are much more useful, particularly in the *regiospecific* synthesis of alkenes, because the double bond is fixed in a single position determined entirely by the starting compounds in the reaction.

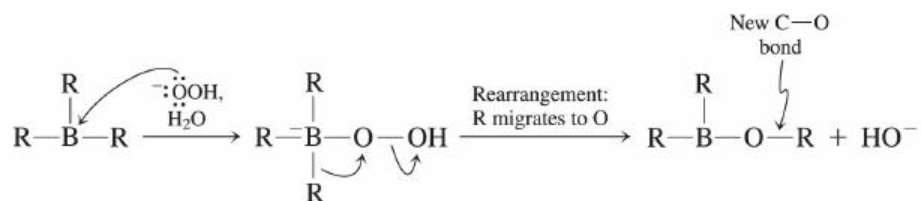


17-13. Baeyer-Villiger Oxidation of Ketones

Baeyer-Villiger oxidation of ketones to carboxylic esters is significant for two reasons. Mechanistically, it involves a nucleophilic addition to a carbonyl group, which is no big deal. The nucleophile, however, is a peroxidic species, which can lead to rearrangement, forming a new oxygen – carbon bond.



You might recall a somewhat related mechanistic rearrangement, the migration of a group from boron to oxygen during the oxidation of alkylboranes with basic hydrogen peroxide.



The second reason the Baeyer-Villiger reaction is significant is that it *cleaves a carbon – carbon bond*. You have seen only a very small number of reactions capable of this (e.g., ozonolysis; [Chapter 12](#)). The power of this method to be a synthetic tool lies both in its high selectivity when the carbonyl compound is not symmetrical (see discussion of migratory aptitudes) and in the subsequent chemistry available from the ester or acid, which will be covered shortly.